

Enterprise Analytics Pipeline using AdventureWorks Database

Project Documentation

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1. Introduction

This project develops a complete analytics pipeline using the AdventureWorks database in PostgreSQL.

The objective is to convert raw transactional business data into a reusable analytics layer that supports reporting, dashboards, and business decision-making.

The project follows a layered architecture where raw tables are accessed only once. All remaining analytics tables are created using previously generated analytics tables.

2. Project Objectives

The main objectives of this project are:

- Build an analytics schema
- Create reusable analytics tables
- Calculate business KPIs
- Perform customer segmentation
- Analyze regional sales
- Evaluate product performance
- Monitor inventory
- Analyze purchasing performance
- Create dashboard-ready datasets
- Demonstrate advanced SQL concepts

3. Project Overview

The pipeline contains **8 logical stages**.

Stage	Purpose
Stage 1	Base Analytics
Stage 2	Business Metrics
Stage 3	Customer Segmentation
Stage 4	Regional Analysis
Stage 5	Inventory & Purchasing
Stage 6	Executive Dashboard
Stage 7	Advanced SQL
Stage 8	Dashboard Datasets

Each stage depends only on tables created in earlier stages.

4. Database Overview

Database Used:

AdventureWorks

Main Schemas Used:

- Sales
- Production
- Purchasing

Major business entities include:

- Customers
- Products
- Orders
- Salespersons
- Vendors
- Inventory
- Territories

5. Analytics Pipeline Architecture

The project follows a layered architecture.

Raw AdventureWorks Tables



Stage 1

Base Analytics Tables



Stage 2

Business Metrics



Stage 3

Customer Segmentation



Stage 4

Regional Analysis



Stage 5

Inventory & Purchasing

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Stage 6

Executive Dashboard

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Stage 7

Advanced SQL Showcase

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Stage 8

Dashboard Extraction Tables

6. Technologies Used

- PostgreSQL
- pgAdmin
- SQL
- Python
- Pandas
- SQLAlchemy
- Matplotlib
- Jupyter Notebook

7. Analytics Schema Design

A separate schema named **analytics** was created.

Advantages:

- Keeps analytical tables separate from operational data.
- Prevents accidental modification of source tables.
- Makes the pipeline organized and reusable.

8. Stage 1 – Base Analytics

Nine base analytics tables were created.

Table	Purpose
sales_analytics	Sales fact table
customer_analytics	Customer summary
product_analytics	Product information
employee_analytics	Salesperson summary
territory_analytics	Territory summary
vendor_analytics	Vendor information
inventory_analytics	Inventory status
purchasing_analytics	Purchasing information
date_analytics	Monthly sales summary

These tables read data directly from AdventureWorks.

9. Stage 2 – Business Metrics

Business performance metrics were calculated.

Tables created:

- monthly_revenue
- quarterly_revenue
- yearly_sales_growth
- product_performance
- salesperson_performance

Metrics include:

- Revenue
- Growth %
- Profit
- Rankings
- Moving averages
- Revenue contribution

10. Stage 3 – Customer Segmentation

Customers were classified into:

- VIP
- Regular
- One-Time Buyer
- Low Value
- No Purchases

Additional tables:

- customer_ltv
- customer_retention
- repeat_customers

11. Stage 4 – Regional Analysis

Regional performance was analyzed.

Created tables:

- regional_performance
- regional_monthly_trend
- regional_growth

Metrics include:

- Territory revenue
- Customer count
- Territory ranking
- Year-over-year growth

12. Stage 5 – Inventory & Purchasing

Inventory analysis includes:

- Low stock products
- Stock health
- Overstock detection

Purchasing analysis includes:

- Supplier performance

- Purchase trends
- Reject rate
- Total spend

13. Stage 6 – Executive KPI Dashboard

A single dashboard table summarizes important KPIs.

KPIs include:

- Total revenue
- Monthly growth
- Yearly growth
- Total customers
- VIP customers
- Average customer lifetime value
- Top product
- Top territory
- Best supplier

14. Stage 7 – Advanced SQL Showcase

This stage demonstrates advanced SQL techniques.

Concepts used:

- Common Table Expressions (CTEs)
- Window Functions
- Ranking Functions
- Conditional Aggregation
- CASE statements
- Multiple joins

15. Stage 8 – Dashboard Extraction Tables

Final reporting tables were created.

These include:

- best_selling_products
- lowest_performing_products

- category_performance
- product_rankings
- top_territories
- lowest_performing_territories

These tables are directly used in the Python dashboard.

16. SQL Features Used

The project demonstrates:

- CREATE TABLE AS
- CTE
- CASE
- JOIN
- LEFT JOIN
- GROUP BY
- ORDER BY
- Window Functions
- RANK()
- NTILE()
- LAG()
- SUM() OVER()
- AVG() OVER()
- FILTER
- DATE_TRUNC()
- CREATE INDEX

17. Performance Optimization

The following optimizations were applied:

- Separate analytics schema
- Reusable analytics tables
- Indexes on frequently used columns
- Raw tables accessed only once

- Layered pipeline design

18. Intermediate Tables Created

A total of **33 analytics tables** were created.

Stage 1

- sales_analytics
- customer_analytics
- product_analytics
- employee_analytics
- territory_analytics
- vendor_analytics
- inventory_analytics
- purchasing_analytics
- date_analytics

Stage 2

- monthly_revenue
- quarterly_revenue
- yearly_sales_growth
- product_performance
- salesperson_performance

Stage 3

- customer_segments
- customer_ltv
- customer_retention
- repeat_customers

Stage 4

- regional_performance
- regional_monthly_trend
- regional_growth

Stage 5

- low_stock_products
- supplier_performance
- purchasing_trends

Stage 6

- executive_kpi_dashboard

Stage 7

- advanced_sql_showcase

Stage 8

- best_selling_products
- lowest_performing_products
- category_performance
- product_rankings
- top_territories
- lowest_performing_territories

19. Dashboard & Visualization

Python dashboards were created using:

- Monthly Revenue
- Revenue by Territory
- Customer Segments
- Top Products
- Category Revenue
- Salesperson Performance
- Inventory Status
- Executive KPI Dashboard
- Quarterly Revenue
- Product Profitability

20. Business Insights

Major insights obtained include:

- Revenue trends over time

- Best-performing territories
- Most profitable products
- Customer purchasing behavior
- Inventory shortages
- Supplier quality
- Salesperson rankings
- Product category performance

21. Challenges Faced

- Designing a reusable analytics pipeline.
- Avoiding repeated access to raw tables.
- Managing dependencies between analytics tables.
- Optimizing SQL queries using indexes.
- Creating meaningful business KPIs.
- Preparing dashboard-ready datasets.

22. Assumptions Made

- AdventureWorks data is complete and reliable.
- Net sales represent actual sales revenue.
- Estimated profit is calculated using standard cost.
- Customer lifetime value is based on total revenue.
- Stock status is determined using reorder point and safety stock level.
- Rankings are calculated using available transactional data.

23. Project Outcomes

The project successfully delivers:

- Reusable analytics schema
- Business metrics pipeline
- Executive KPI dashboard
- Customer segmentation
- Regional analysis
- Inventory analytics

- Purchasing analytics
- Dashboard-ready datasets
- Advanced SQL implementation

24. Future Improvements

Possible future enhancements include:

- Real-time analytics
- Materialized views
- Power BI integration
- Interactive dashboards
- Predictive analytics using Machine Learning
- Automated ETL pipeline

25. Conclusion

This project successfully transforms AdventureWorks operational data into a structured analytics layer using PostgreSQL. The layered architecture ensures maintainability, scalability, and efficient query execution while supporting executive dashboards, business reporting, and data-driven decision-making. The implementation demonstrates practical use of SQL analytics, indexing, window functions, CTEs, customer segmentation, and KPI reporting, making it a strong enterprise-level analytics solution suitable for reporting and visualization.